### 3.1.3 The development of species: evolution and classification

| | Learning outcomes | Additional guidance |
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| | *Learners should be able to demonstrate and apply their knowledge and understanding of:* | |
| **(a)** | the concepts of biological classification and species | To include the taxonomic ranks in the hierarchy of classification (domain, kingdom, phylum, class, order, family, genus, species)<br>**AND**<br>an outline of the biological and phylogenetic species concepts. |
| **(b)** | the types of evidence used in biological classification and consideration of how theories change as new evidence is found | Evidence for hominid classification to include observable features (e.g. fossils) and molecular evidence (e.g. DNA).<br>HSW7, HSW8 |
| **(c)** | the use of DNA barcoding in biological classification, examples of the genes used and consideration of the reasons for the choice of these genes | To include the use of mitochondrial genes (e.g. cytochrome c oxidase 1) in animals, and chloroplast genes in plants (no details of electrophoresis are required). |
| **(d)** | the interpretation of phylogenetic trees and genetic data to show relatedness and classification in plants and animals | To include consideration of hominids, both extinct and extant, and hylobatids, including examples in which there is conflicting evidence.<br>HSW5, HSW6, HSW8 |
| **(e)** | **(i)** behavioural, physiological and anatomical adaptations to the environment<br><br>**(ii)** practical investigation into adaptations of plants to environmental factors | To include the following adaptations in *Homo sapiens*: tool use and cultural adaptations for social bonding (behavioural), lactose tolerance and skin pigmentation (physiological), bipedalism and brain size (anatomical)<br>**AND**<br>adaptations of plants to their environment including adaptations to extremes of temperature, light and water. |

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| **(f)** | the evolution of language as an example of a scientific question with many competing theories | To include discussion of why some scientific questions (e.g. “how did language evolve?”) are difficult to answer because of a lack of evidence, and consideration of competing theories (to include the “mother tongues” and “gossip” hypotheses).<br>HSW1, HSW2, HSW3, HSW7, HSW8 |
| **(g)** | adaptation and selection as components of evolution | To include the ideas of genetic variation, selection pressures and natural selection in relation to evolution. |
| **(h)** | the definition and measurement of biodiversity | To include calculating Simpson’s Index of Diversity ($D$). The formula will be provided where needed in assessments and does not need to be recalled<br>$D = 1 - \left( \Sigma \left( \frac{n}{N} \right)^2 \right)$<br>To include a consideration that biodiversity can exist at the genetic, species and ecosystem levels.<br>*M0.2, M0.3, M0.4, M1.1, M1.5, M2.3* |
| **(i)** | the calculations of genetic diversity within populations. | To include the percentage of gene variants (alleles) in a genome.<br>$\text{proportion of polymorphic gene loci} = \frac{\text{number of polymorphic gene loci}}{\text{total number of loci}}$<br>*M0.1, M0.2, M1.1, M2.2, M2.4* |